姓名：

学号：

学院：

上海科技大学

2016-2017 学年第 1 学期本科生期末考试卷

开课单位：上海科技大学生命科学与技术学院

授课教师：廖军，陈佳

考试科目：**Biochemistry I**

课程代码：BIO1003

考生须知：

1. 请严格遵守考场纪律，禁止任何形式的作弊行为。
2. 参加闭卷考试的考生，除携带必要考试用具外，书籍、笔记、掌上电脑和其他电子设备等物品一律按要求放在指定位置。
3. 参加开卷考试的考生，可以携带教师指定的材料独立完成考试，但不准相互讨论，不准交换材料。

考试成绩录入表：

题目	1	2	3	4	5	6	7	8	9	10	总分
计分											
复核											

评卷人签名：　　复核人签名：

日期：　　日期：

编写说明：

1. 要求评卷人和复核人不能是同一人。
2. 试卷内页和答题纸编排格式由各学院和出题教师根据实际需要自定，每页须按顺序标注页码（除封面外），要求排版清晰、美观，便于在页面左侧装订。为方便印刷归档，建议使用 A4 双面印刷（学校有印刷一体机提供）。
3. 主考教师编写试卷时尽可能保证试题科学、准确、合理，如考试过程中发现试题有误，主考教师需负责现场解释，此类情况学校将作为教学评估记录的一部分。

Multiple choices or single choice questions

1) Signal sequences are part of a protein that (C)
A. signal folding of the protein
B. signal the protein synthesis on the ribosomes is ended
C. transport proteins to other sites within the cell
D. refold proteins in prion-associated diseases

2) Beta pleated sheets are examples of protein's (B)
A. primary structure
B. secondary structure
C. tertiary structure
D. quaternary structure

3) Protein folding is (B)
A. automatic, mediated by the protein itself
B. mediated by other proteins called chaperones
C. mediated by the ribosomes
D. none of the above

4) ADP contains? (A)
A. furanose ring.
B. two phosphoandydride bonds.
C. a β-O-glycosidic bond.
D. a pyrimidine base.

15) In living cells, various nucleotides and polynucleotides: (AB(D))
A. serve as structural components of enzyme cofactors.
B. alter the equilibria of chemical reactions.
C. carry metabolic energy
D. serve as intracellular signaling molecules

6) Plasmids are: (D)
A. a part of the bacterial chromosome.
B. found only in mitochondria.
C. composed of RNA and protein.
D. closed circular DNA molecules.

7) Both general recombination and transposon-type recombination: (D)
A. occur between homologous regions on chromosomes.
B. are important in the repair of damaged DNA.
C. involve RecA protein.
D. generate genetic diversity.

8) The oligomer of four subunits of the hemoglobin (Hb) gene represent protein's (D)

A. primary structure
B. secondary structure
C. tertiary structure
D. quaternary structure

9) Polyprotic acids such as H_3PO_4, can act as acid-base buffers (D)

A. only in combination with polyprotic bases
B. if their concentration is kept low
C. at pH values around neutrality
D. at pH values around any of their pK_a's

10) When the amino acid alanine (R-group is CH_3) is added to a solution with a pH of 7.3, alanine becomes (C)

A. a cation
B. nonpolar
C. a zwitterions
D. an isotope

11) The α-amino acids have a carboxyl group with a pK around _____, and an amino group with a pK near _____. (D)

A. 1.1, and 12.1
B. 6.5, and 8.0
C. 3.3, and 10.5
D. 2.2, and 9.4

12) In the laboratory, some *E.coli* cells have been grown with transcription inhibitors so that only replication could occur. You break open the cells; gently purify the DNA, separating it from the rest of the cell contents; then treat the intact DNA with radioactively labeled antibodies to DNA ligase and DNA polymerase I. Where do you expect to find the highest concentration of these antibodies? (D)

A. On the leading strand, in one direction from the point of origin.
B. On the leading strands, in both directions from the point of origin.
C. On the lagging strand, in one direction from the point of origin.
D. On the lagging strands, in both directions from the point of origin.

13) On Mars, it seems, the genetic material is double-stranded RNA rather than DNA. Martian RNA polymerase uses the double-stranded RNA as template to make mRNA; it is an RNA-directed RNA polymerase. Assume all other aspects of transcription and translation and the functions and integrity of the enzymes involved are the same on Mars as on Earth (i.e., RNA has the same function, the

same base pairs are complementary to each other, and chemistry in general is the same). Over time, the integrity of the genetic message in Martian RNA would: (C)

A. decline more rapidly than that of DNA because of the lack of editing function of RNA polymerase.
B. be comparable with that of DNA.
C. decline more rapidly than that of DNA because the deamination of cytosine to uracil in RNA would confuse the message.
D. be greater than that of DNA because the message would be more direct: RNA to protein, with one less step in which mistakes could happen.

14) Which of the following event(s) can occur during the processing of eukaryotic mRNA transcripts? (CD)

A. Attachment of a poly (A) tail to the 5' end of the transcript.
B. Methylation of all G residues.
C. Excision of introns.
D. Alternative splicing to produce two different proteins.

15) The unfolding of regular secondary structure causes (D)

A. little increase in the entropy of protein
B. large decrease in the entropy of the protein
C. no change in the entropy of the protein
D. large increase in the entropy of the protein

16) The genetic code is degenerated. Which of the following codons represents the principle of degeneracy? (CD)

A. UAA and UAC
B. AUG and AUA
C. CAU and CAC
D. UUA and UUC

17) The σ^{70} subunit of the *E. Coli* RNA polymerase: (B)

A. acts as the catalytic site for polymerization.
B. recognize promoters.
C. has a proofreading function.
D. increases the processivity of the enzyme.

18) Following the 5' to 3' convention of writing nucleotide sequences, indicate which of the following mRNA codon(s) can be recognized by the tRNA anticode ICG. (A) GC

A. UGC
B. CGA
C. UGA
D. CGU

19) Which of the following statement(s) about the *lac* operon is (are) true?
(CD)

A. All seven genes are transcribed into mRNA, which is then translated into five different proteins.
B. The repressor binds to the structural genes, preventing their transcription.
C. Highly efficient binding of RNA polymerase to DNA occurs only in the presence of CRP and cAMP.
D. In the absence of lactose, the repressor binds the operator.

20) Each tRNA molecule has a cloverleaf secondary structure consisting of
(A)

A. three stem loops
B. two stem loops
C. four stem loops
D. either (a) or (b)

Fill-in the blank questions

1) 变构 is any change in the shape or conformation of the protein from the native state.

2) The bond which joins one monosaccharide to another to form a disaccharide is called a 糖苷 bond.

3) Name the polysaccharide which is found in humans which is similar to starch found in plants. 糖原

4) During 翻译, the information encoded in the base sequence of each mRNA is converted to the corresponding sequence of amino acids. This process has three main parts: 起始, 延长 and 终止.

5) Ribosomes are the site of 蛋白质合成 Ribosomes are composed of 2 subunits. Ribosomes are made up of rRNA and 蛋白质.

6) Uracil is a nitrogenous base found predominantly in 核糖核苷的 acids.

7) Thymidylate is a nucleotide found predominantly in 脱氧核苷的 acids.

8) PCR is the abbreviation of 聚合酶链式反应

9) The new, improved product of two fused genes is a 重组 protein.

10) DNA that is not supercoiled and not underwound is said to be in a 放松的 state.

11) Monosaccharides, disaccharides, and polysaccharides all belong to a class of organic molecules known as 糖（碳水化合物）.

12) During protein synthesis, amino acids are brought to ribosomes by RNA molecules known as tRNA. One 氨酰tRNA attached to its specific amino acid binds by a special sequence of three nucleotides, known as 反密码子, to an mRNA codon exposed in the ribosome.

13) Those first amino acids of the newly formed protein molecules that act to "direct and inject" the entry of the protein into the cisterna of the endoplasmic reticulum, are known as the 信号肽.

14) Carbohydrates are the main fuel molecules for cellular work. Carbohydrates are a class of molecules ranging from the simplest sugars, called 单糖, to giant molecules called 多糖, built of many sugars.

15) The enzyme that catalyzes formation of a phosphodiester bond between a 3'-hydroxyl at the end of one DNA strand and a 5'-phosphate at the end of another is DNA连接酶.

16) The enzyme that unwinds DNA is 解旋酶.

17) 嘧啶二聚体 are the DNA lesions repaired directly or by ABC excinuclease in nucleotide-excision repair.

18) In the liver, the fructose is converted to glucose. This process is relatively easy because glucose and fructose are 立体异构体 having the same molecular formula, written C6H1206, but slightly different structures. Glucose circulates around the body as "blood sugar" and is taken up by the cells for fuel as needed. Extra glucose molecules are taken up by liver and muscle cells and linked together by α1-4糖苷键 synthesis to form a polysaccharide called 糖原.

19) Proteins necessary for the activity of RNA Pol II are 转录 factors.

20) RNA-dependent DNA polymerase is 逆转录酶.

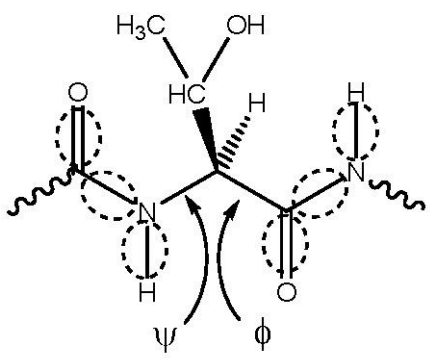
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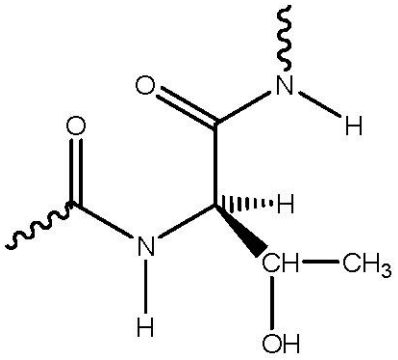
Cumulative

1) As a teaching assistant for a biochemistry course, you are expected to grade exams. The question that you have been assigned to grade asks is two parts. In the first part, students must complete a portion of a polypeptide chain (left) with:

A. sidechain for L-tyrosine
B. Circles on bonds about which there is no rotation
C. Arrows indicating the bonds about which ϕ and ψ conformational angles are measured.

In the second part (right), students are asked to draw the same peptide segment with ϕ =180° and ψ=0°. One student gives the following answers:





What is wrong with these drawings? Briefly list up to 6 different errors, one per line, (there may actually be more or fewer), that you see in the students' answers. (Include whether the error is in the left structure, right structure, or both)

2) How can UV light induce mutagenesis of *E. coli* DNA and why should the mutagenesis of *E. coli* using UV radiation be done in a laboratory without windows?

有两方面原因:
(1)阳光中有长波段的紫外线,如果透过窗进入实验室,很可能会对实验中施加的定量紫外线辐射产生干扰而引入其他的变量,使得大肠杆菌诱变结果和实验设计时的紫外线辐射强度(自变量)之间的关联性降低、诱变结论可信度下降.
(2)此外,进入的阳光由于含有300-500nm的蓝色波段,会激活细菌中的MTHFpolyGlu分子,启动DNA的直接修复机制,使得实验中先前紫外线辐射下引起的嘧啶二聚体被直接修复(边诱变边修复).这样就会造成紫外线诱变结果被改变或消除,而在诱变后无法被有效检测到,从而使得UV诱变效果的结论偏离实际情况.

3) In eukaryotes, gene transcription requires a number of genetic elements and associated binding proteins. For example, some genes have a TATA box that is located 25-30 base pairs upstream from the transcription start site and must be positioned on the 5' side of the gene. In addition there are enhancer elements that are usually located 100 to 5,000 base pairs away from the start site and can be located on either the 3' or the 5' side. Explain this discrepancy.

差异解释:

(1)TATA Box是基础转录的起始位点.转录开始时,各个基础转录因子和RNA聚合酶II按照一定的次序,通过招募的方式形成转录前起始复合物(PIC),结合在TATA Box上.为了让RNA聚合酶II成功识别启动子序列,从而启动基础转录,TATA Box的位点必须位于表达基因的上游(5')且距离启动子(转录起始位点)较近(即题目中25-30bps)的位置;

(2)增强子是真核生物中激活蛋白(一种转录因子)和DNA的结合位点,经过招募激活蛋白和一系列的辅激活蛋白形成增强体,并将组蛋白修饰酶、染色质重塑因子和基础转录因子有序地招募到启动子周围,从而激活基因的表达.在这个过程中,增强子和转录起始位点之间的HMG蛋白可以通过弯曲DNA链从而将增强体及其上面的转录因子移到转录起始位点的旁侧,使其可以促进转录的起始.因此,HMG蛋白的作用使得增强子序列不必位于转录起始位点的周围,可以相距较远(即题目中100-5000bps),也可以在转录起始位点的上游(5')或下游(3'),即两侧均可以实现基因表达的激活.

4) Many metabolic reactions are coupled reactions. A common coupled reaction is substrate phosphorylation by ATP. For example, the oxidation of glucose begins with its phosphorylation to glucose-6-phosphate and is coupled to ATP hydrolysis.

$$\text{Glucose} + \text{Pi} \longleftrightarrow \text{Glucose-6-phosphate} + H_2O \qquad \Delta G^{0\prime} = 14\text{k J/m ol}$$

$$\text{ATP} + H_2O \longleftrightarrow \text{ADP} + \text{Pi} \qquad \Delta G^{0\prime} = -30.5\text{kJ/mol}$$

In muscle cells at 37° C, the steady state ratio of [ATP]/[ADP] is 12. Assuming that glucose and glucose-6-phosphate (G6P) achieve equilibrium values in muscle, what is the ratio of [G6P] to [glucose]?

delta G=-RTlnK!